Project Update (AI)

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Successes

- Very helpful for initial CFD learning
    - walkthrough of a SimScale setup and quizzes on CFD concepts, uses, limits, and reliability
- Quickly made SimScale documentation useful for broad CFD/SimScale tips
- Fusion AI for CAD produced a good VJ20 model in about four minutes, under one-third the time ZooKeeper took, with comparable or better initial quality
- AI-assisted research identified a published NACA 0012 validation approach that motivated a 2D model; this substantially improved the reported lift result
- LLM summarized daily documentation and CFD iterations and pulled out important events and themes

Failures

- Quickly began repeating known information and stopped providing valuable diagnoses for the airfoil-validation problem
    - did not reliably track which CFD settings had changed per simulation run
- LLM misreported top design cut-in speed, assigning a value from a different design (also mentioned in that paper)

Questions

- How should we go about validating quantitative data before it goes into decision making? human spot-checks, a second-model check, schema constraints, or automated source-to-claim links?
- How can we distinguish a model limitation from a retrieval/source-coverage problem or prompt issue when the wiki provides repetitive responses?
- What retrieval setup could we use to let the wiki use selected sources *and* external CFD sources while still being trustworthy? Is it worth it to pursue this?